



# Implementation of micro-credentials in higher education: A systematic literature review

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## Abstract

Micro-credential (MC) is an innovative and disruptive technology-based learning model. In higher education (HE), MCs are typically certified in the form of a digital badge (DB) issued to learners upon completion of a credential. The introduction of MCs and DBs has the potential to revolutionise the way in which HE institutions deliver degrees, and how students learn and achieve their tertiary qualifications. To explore this innovative learning, this study investigates the potentials and challenges of implementing MCs in HE. Through a systematic literature review of 56 peer-reviewed articles on MCs and related DBs in HE published between 2015 and 2021, we find that research on MCs in HE is gaining attention. While different aspects of MCs in HE have been studied using various research methodologies and theoretical lenses, further research is needed to capture the influence of technology and to understand different stakeholder perspectives. This study contributes to the growing literature on MCs in HE by synthesising the relevant knowledge areas into key themes and developing a conceptual framework for implementing MCs in HE. This study also identifies critical research gaps to advance the literature on MC implementation in HE.

**Keywords** Micro-credential (MC) · Digital badge (DB) · Higher education (HE) · Digital certification

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## 1 Introduction

Micro-credentials (MCs) are competency-based learning models provided by higher education (HE) or business which issue learners with a digital badge (DB) upon completion (Alamri et al., 2021). DBs are therefore a specific form of MC, providing a visual representation of MC completion and a proof of learning or evidence of acquired skills (European Commission, 2020; Oliver, 2019).

Introducing MCs in HE holds the potential to change how HE institutions offer degree programs and students acquire their qualifications (Greene, 2019; Lockley et al., 2016). A relatively new technology-based learning concept, MCs are considered innovative pedagogical tools (Newby & Cheng, 2020) and are fast gaining recognition in the HE landscape, being seen as one of the disruptive forces of change under Higher Education 4.0 (Brown et al., 2021).

Shorter than traditional forms of accredited learning (Brown et al., 2021), MCs may be an additional, alternate, complementary or component part of a formal qualification (Oliver, 2019). The Association to Advance Collegiate Schools of Business defines MCs as certifications granted by assessed mastery of a specialised competency (AACSB, 2021). MCs, as such, may be considered as a sub-unit of a credential that may accumulate into a larger credential or degree or be part of a learner's portfolio (Greene, 2019; MicroHe, 2017; Oliver, 2019). Importantly, MCs have to align with formal qualification levels such as a Bachelor or Master's degree (Oliver, 2019).

The global online degree market with MCs is predicted to grow exponentially. In 2019, individuals spent about US \$36 billion on online degrees of which approximately US \$9.8 billion were expended on MCs (HolonIQ, 2021b). The value of online degrees is expected to grow to US \$74 billion by 2025, where MCs will have a significant share (HolonIQ, 2021a). Industry 4.0 technologies have reshaped the skill requirements of jobs, which requires re-training, reskilling or redeployment of the workforce (Brown et al., 2021). To reposition their competitive edge, many global companies have introduced credentialed short courses which can be recognised as equivalent to a full Bachelor's degree for recruitment purposes (Brown et al., 2021). Likewise, universities, HE quality assurance authorities, and government policymakers are also placing greater importance on introducing MCs in HE. Already, many HE institutions are independently working on introducing MCs into their degree programs (Wheelahan & Moodie, 2021a). HE regulatory bodies are also working to standardise and to regulate MCs offerings across institutions and regions (Brown et al., 2021).

Likewise, the literature on MCs and DBs in HE has been developing quickly in recent years. Our literature search indicates that at least five literature review articles have been published in this emerging area between 2016 and 2021. However, they have not focussed explicitly on MCs and DBs in HE. An early integrative review by Mah (2016) studied the nexus between learning analytics, digital badging and student retention. Subsequently, Liyanagunawardena et al. (2017) conducted a systematic review of the literature from 2011–2015, focussing solely on open badges but not specific to DB in HE. Recently, Selvaratnam and Sankey

(2021) conducted a general literature review of MCs in HE, which covers only the Australasian landscape using a mix of academic journals and grey literature, but is not “systematic” by definition (Denyer & Tranfield, 2009; Tranfield et al., 2003). With the growing literature in the field in the last decade, we contend that the cumulative findings and concepts developed in MCs studies have much to offer to advance the notion of introducing MCs in the HE sector. A systematic literature review of studies of MCs in HE can identify research trends, explore advancements in the field and gaps for further studies, and lay the foundation for building a theoretical base of the key knowledge areas of MCs in the HE sector. On this note, this systematic review study aims to answer the following research questions (RQ):

- (1) What are the recent trends of MCs research publications in HE?
- (2) What are the foci, key knowledge areas and clusters of existing MC research in HE?
- (3) What are the potentials and challenges for the implementation of MCs in HE?
- (4) What are the critical research gaps and the opportunities available for further investigation?

The rest of the paper is organised as follows. Section 2 provides a review of the background literature and Sect. 3 describes the research methodology. Section 4.1 analyses the recent trends of MCs research publications in HE (RQ1) and Sect. 4.2 conducts thematic analysis and presents critical reflection of key knowledge areas of MCs research in HE (RQ2). Through a conceptual framework, Sect. 5 provides insights into the potentials and challenges of implementing MCs in HE (RQ3). Section 6 summarises critical research gaps and outlines future research directions (RQ4) and Sect. 7 concludes the study.

## 2 Research background

### 2.1 Micro-credentials and digital badges in higher education

Micro-credentialing is a means of certifying learning accomplishments smaller than that of a full degree (Selvaratnam & Sankey, 2021) and symbolises a proof of learning, where learning outcomes have been assessed against transparent standards (European\_Commission, 2020).

MCs may be offered to provide hard skills associated with technical course work or to improve day-to-day interaction skills and awarded for successful completion of specific components or competencies of a course or program (Eraut, 2012). MCs vary in size and may be a micro- or sub-unit of a credential or qualification (European\_Commission, 2020). For example, under the European Credit Transfer System (MicroHe, 2019), MCs must confer a minimum of five credits. The New Zealand Qualifications Authority (NZQA, 2019), however, define MCs as between five and 40 credits in size.

The growing trend in the use of MCs in HE is indicative of an emerging shift towards competency-based education (Blackburn et al., 2016). MCs enable individuals to upskill or reskill knowledge and capabilities that are in high demand in the labour market, and to prove and share those accomplishments with employers and other parties (European\_Commission, 2020).

MCs encompass various forms of certification, including nano-degrees, micro-masters, credentials, certificates, badges, and endorsements offered by HE institutions (EU, 2022; Malczyk, 2019). Among them, DB has emerged in recent years as a popular form of MC (Pitt et al., 2019). In the HE sector, MCs are issued in the form of DBs, as a digital proof of learning, skills and competencies (Kukkonen, 2021). DBs in HE are awarded for participation in activities towards the completion of a MC, or upon completion of a MC (Abramovich, 2016; Eager & Cook, 2020; Fanfarelli & McDaniel, 2017). DBs have been introduced as an alternative to grades, which may be “earned through an automated process following the successful completion of key learning activities” (Garnett & Button, 2018, p. 2). DBs represent an accomplished skill or knowledge and comprise a variety of metadata (including learning requirements, instructional materials, endorsement information, issue data and institution) which allows the badge to be created, acquired and shared in an online space (Cheng et al., 2020).

In 2007, James Gee, a professor of literacy studies at Arizona State University, first pitched the DB idea, comparable to badges used in video games to indicate progress, as an alternative credential or qualification (Gibson et al., 2015; Grant, 2016). In 2012, the Harvard Business Review reported DBs as one of four innovative future trends to watch (Schrage, 2012). DBs have given rise to a global discussion on educational practices and possibilities centred on evidence-based learning and assessment (Alt, 2021). Internationally, the use of DBs in HE is increasing (Coleman, 2018; Hartnett, 2021).

The terms MC and DB are often used interchangeably (Clayton et al., 2014). MCs and DBs are awarded by business organisations or HE institutions after the acquisition of certain knowledge or demonstration of competence to ensure authenticity and criteria of quality (European\_Commission, 2020; Fanfarelli & McDaniel, 2017). Therefore, DBs can be considered as a specific form of micro-credentialing (Dyjur & Lindstrom, 2017; Ippoliti, 2014). Thus, the scope of this study includes MCs, and DBs that are essentially MCs or are used as MCs.

## 2.2 Micro-credential review articles

Table 1 shows the key review studies on MCs undertaken until 2021. The earliest review by Mah (2016) draws on information and literature related to MCs as of 2015 and does not follow a systematic literature review format, such as that prescribed by Tranfield et al. (2003). The review by Liyanagunawardena et al. (2017) includes articles published between 2011 and 2015 but does not have a specific HE focus and is limited in scope to open badges. A systematic review of DBs by Noyes et al. (2020), includes 201 articles from 2008 to 2019, and focuses exclusively on health care education. A more recent integrative review by Selvaratnam and Sankey (2021)

**Table 1** Notable MC review articles

| Review article                  | Scope/topic                                                                                                             | Review type and database                                                                                                                                                      | Coverage                                                                                                           | Key focus and outcomes                                                                                             |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Alamri et al. (2021)            | Review of theory and practices of personalised learning in HE; DBs, competency-based learning, and adaptive learning    | Systematic search of online libraries and databases such as EBSCO, Elsevier, ProQuest, Web of Science, Social Sciences Citation Index, Taylor & Francis Group, Google Scholar | 84 articles (peer reviewed and non-peer reviewed), of which only 15 are related to DBs and personal learning       | DBs barriers, challenges, and theoretical and practical implications of implementing a personalized learning in HE |
| Selvaratnam and Sankey (2021)   | Review of studies on MC implementation in HE in Australasia                                                             | Descriptive analysis of academic and non-academic articles (such as media and industry articles)                                                                              | 20 academic articles between 2016–2019; many non-peer reviewed academic articles, reports, whitepapers             | Limited to the Australasian landscape                                                                              |
| Noyes et al. (2020)             | Review of how DBs have been used in the context of health care education programmes                                     | Literature review of academic and professional medical research using MEDLINE, Web of Science, CAB Abstracts and ScienceDirect                                                | 201 medical literature from January 2008 to March 2019                                                             | Design and implementation of DBs within medical education as a learner-centred and competency model                |
| Liyanagunawardena et al. (2017) | Review on open badges, not specific to the HE sector                                                                    | Systematic review of the MC literature from sources/database such as Scopus, Web of Science, and Google Scholar                                                               | 135 articles between 2011–2015 (53 journal papers, 76 conference papers, five book chapters & one research report) | Review literature was limited to open badges                                                                       |
| Mah (2016)                      | Review of the interconnectedness of digital badging, academic competencies and learning analytics for student retention | Integrative literature review of theoretical literature as well as current practices and experience; database not mentioned                                                   | No systematic steps were followed                                                                                  | Focuses on learning analytics and DBs with limited coverage of the literature                                      |

of 20 articles from 2016 to 2019 concentrates only on the Australasian region and includes reports, whitepapers, empirical research, conceptual papers, and reviews. Another recent integrative review by Alamri et al. (2021) covers 84 articles and looks broadly at personalised learning approaches and the technology models that support these approaches within blended learning environments in HE. The 84 articles comprise a mix of peer reviewed and non-peer reviewed articles, of which only 15 relate to DBs.

In sum, extant review studies on MCs either have not followed a widely acknowledged reproducible systematic review process (e.g., Tranfield et al. (2003)), including limiting the review to peer-reviewed scholarly articles or are narrow in scope and provide limited coverage of the MC and DB in HE literature (focusing only on gamification, learners' behaviour, personalised learning, Australasian context, health care education, open badges, or learning analytics). To effectively integrate MCs into HE offerings, a systematic review of MCs and DBs in HE articles would make a worthwhile contribution to the literature.

### 3 Research design

We adopted the rigorous guidelines recommended by Tranfield et al. (2003) and Denyer and Tranfield (2009), and Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher et al., 2015), and followed the three-stage process of structured analysis of literature review:

- Planning the reproducible review methodology (design, development, inclusion and elimination criteria, literature search approach),
- data screening, synthesis, and analysis (executing), and
- dissemination (reporting)

#### 3.1 Planning the reproducible review

In a systematic literature review, planning includes identifying the field of literature for review, the unit of analysis, the database(s) to source materials, the keywords to conduct the literature search, the inclusion and exclusion criteria for data clean-up, and the means to store data for analysis and reporting (Ahsan & Rahman, 2022).

As our overarching objective is to investigate research trends of MCs and use of DB for MC in HE, our unit of analysis is a single article focusing explicitly on MCs and DBs related to HE. To search for targeted articles, we relied on the Scopus database, a widely used and reliable source of data for literature reviews (Ahsan et al., 2022). We included all articles available within the database in 'final' or 'article in press' publication stage until December 2021. As a preliminary search criterion, we restricted the search to articles that included keywords present in the title and/or in the abstract. Figure 1 shows the combinations of keywords entered into Scopus: "micro-credentials", "digital badge",

| Keywords                                     | Initial searched articles | Useable articles<br>(conference and non-English articles excluded) | Useable articles<br>(duplicate, out of scope and non-higher education articles excluded) | Total useable articles<br>(Scopus and additional articles from Google Scholar search combined) |
|----------------------------------------------|---------------------------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| 'Microcredential' and 'higher education'     | 8                         | 4                                                                  |                                                                                          |                                                                                                |
| 'digital badge' and 'higher education'       | 76                        | 58                                                                 |                                                                                          |                                                                                                |
| 'micro-credentialing' and 'higher education' | 9                         | 8                                                                  |                                                                                          |                                                                                                |
| 'micro-credential' and 'higher education'    | 28                        | 13                                                                 |                                                                                          |                                                                                                |
| 'digital badge' and 'university'             | 71                        | 44                                                                 |                                                                                          |                                                                                                |
| 'micro-credential' and 'university'          | 29                        | 13                                                                 | 140-60 (duplicate articles) = 80-27(out of scope)=53                                     | 53+3=56                                                                                        |

**Fig. 1** Article selection process of MCs in HE

“higher education” and “university”. These keywords formed the search protocol to identify the literature pertaining to MCs and DBs in HE.

Our initial search identified 221 papers. Discarding non-English language articles and papers published in conference proceedings or in grey literature, such as reports, theses, dissertations, and magazines, we obtained 140 papers. After removing duplicates, 80 articles remained. We read the 80 papers carefully to eliminate ‘loosely related’ articles, i.e., those with content not consistent with the objectives of the study, such as articles focusing broadly on MCs for library use or short courses not specifically on HE, open badges, and papers focusing only on faculty development, teacher training, or professional development. We also excluded articles which were not available in full text within the university library database. Through this exclusion exercise, a total of 27 articles were dropped, leaving 53 relevant articles. To minimise missing work relevant to the review, we went through a similar search process using Google Scholar. Three additional articles (Eager & Cook, 2020; Lemoine & Richardson, 2015; Ralston, 2021) were identified, yielding 56 useable articles for our systematic review. Information on the entire search process is auditable for reviewing and tracking purposes. For each of the 56 selected articles, bibliometric data (bibliographic information, abstract, key words, and references) were extracted for analysis. The bibliometric information of the 56 articles reviewed is provided in Appendix Table 3.

### 3.2 Execution of extracted data

We used VOSviewer, a reputed, reliable and valid software to conduct the co-word analysis (Van Eck & Waltman, 2014) to develop networks of keyword co-occurrences to identify groups of well-connected key words (themes) in MC studies (Callon et al., 1983). In co-word analysis, the unit of analysis is a concept, not a document reference, author, or journal, and it is not related to time function (Ahsan & Rahman, 2022; Zupic & Čater, 2014). This analysis assumes that each field of study can be characterised by a list of keywords and the keywords in each publication can be measured for similarity to show a relationship between two publications (De la Hoz-Correa et al., 2018).

To prepare the content for co-word analysis, we removed formatted abstract titles, copyright information, punctuation marks, and symbols. We amalgamated words conveying similar meaning or concepts (e.g., learner, learners, student and students; undergraduate student and higher education student; credential, credentials, credentialing, micro-credential, micro-credentials and micro credential; universities and institutions; motivation, extrinsic motivation, and intrinsic motivation) to make the analysis more pragmatic (Cobo et al., 2011). As the theme or scope of the research is to group MC knowledge areas, keywords such as ‘journal’, ‘paper’, ‘finding’, ‘research’, ‘methodology’, ‘study’, ‘analysis’, were excluded from the cluster analysis.

The cluster analysis generated four clusters of keywords, each of which is shown in a separate colour (green, red, yellow and blue) in Fig. 2. These clusters are not mutually exclusive since many articles straddle across clusters. In Fig. 2, nodes represent the keyword while links represent the connection between two keywords. The size of the node indicates the co-occurrence frequency of the keywords: the larger the size, the more frequent the keyword co-occurred with another key word.

### 3.3 Reporting

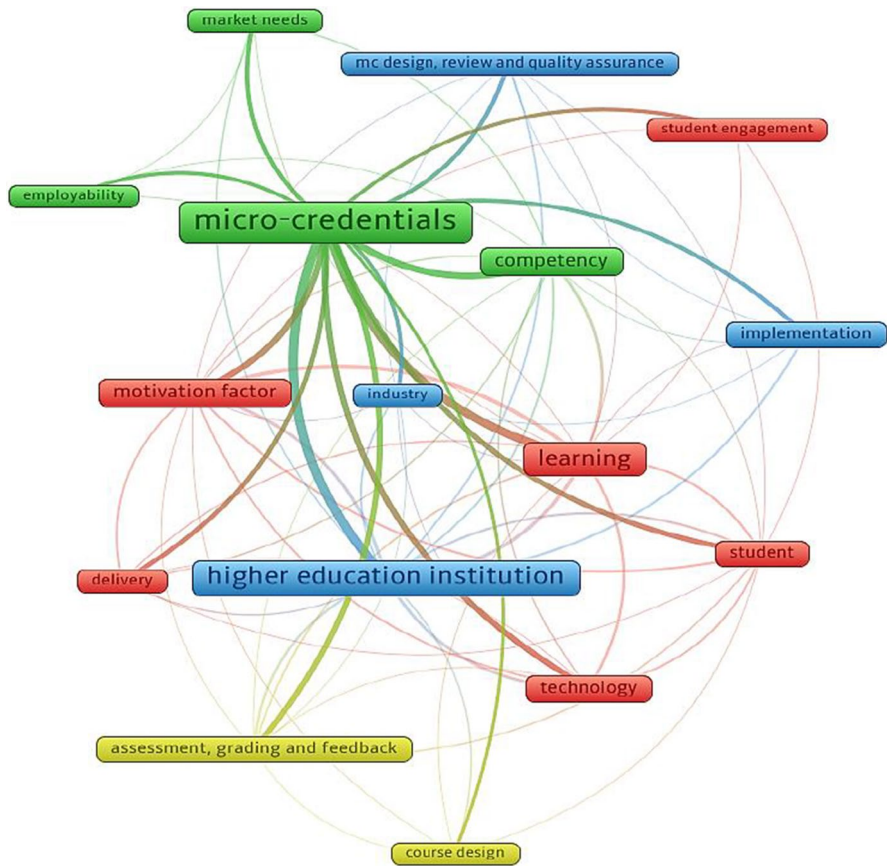
We adopted Tranfield et al. (2003) two-fold reporting approach: descriptive analysis and thematic analysis. Based on bibliometric data (Appendix Table 3), we conducted an in-depth descriptive analysis of the 56 review articles (Sect. 4.1). Using the keyword clusters of Fig. 2, we analysed the underlying themes of MCs in HE research (Sect. 4.2).

## 4 Results and analysis

### 4.1 Descriptive analysis

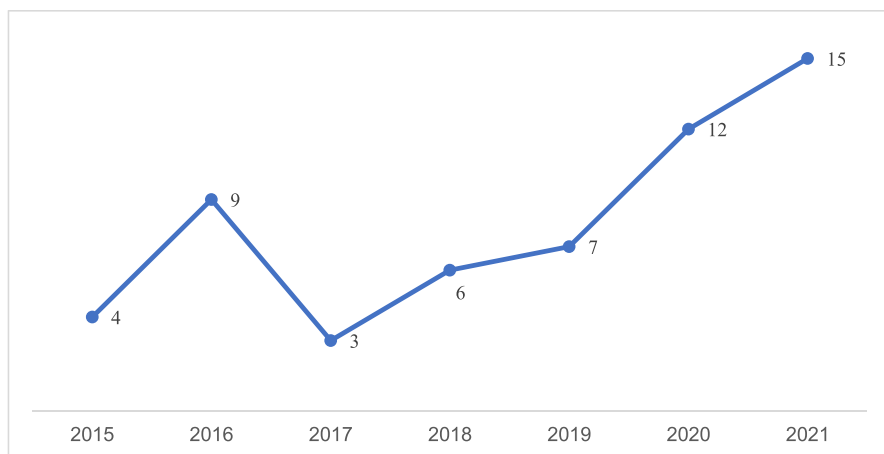
To answer the status of MCs research publications, article data were initially assessed with respect to the number of publications per year, major journal outlets, author, citation, and research methodology for subsequent analysis.





**Fig. 2** Keyword clusters of MCs in HE research from VOSviewer

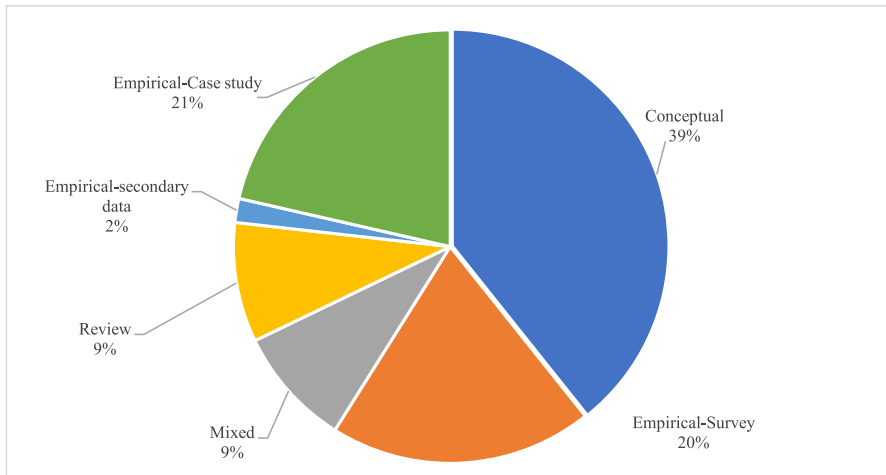
- (i) **Evolution of micro-credentials research:** The 56 articles reviewed were published between 2015 and 2021 (see Fig. 3), of which only 16 (28%) were published before 2017 and the remaining 40 (72%) were published after 2017. Our analysis reveals that MCs research only started to gather steam after 2017 and has been rising since, with 27 articles (48%) published between 2020 and 2021. Our review, as such, not only extends Selvaratnam and Sankey's (2021) study of 20 papers published between 2016 and 2019, but, more significantly, adds richness to the discussion on implementing MCs in HE.
- (ii) **Prominent contributing journals:** The 56 articles were published in 42 journals and two edited books. *TechTrends* was the most popular journal for MCs articles (5 articles), followed by *Education and Information Technologies*, *Educational Technology Research and Development*, *Journal of Teaching and Learning for Graduate Employability*, *Nurse Education and Practice*, *On the Horizon*, and *Journal of Curriculum Studies* (2 articles each). Of these journals, 15 are in the



**Fig. 3** MCs in HE articles published between 2015 and 2021

education domain, and 16 are in education and technology focused journals. The book chapters are from *Foundation of Digital Badges and Micro-Credentials: Demonstrating and Recognizing Knowledge and Competencies* (4 articles), and *Digital Badges in Education: Trends, Issues, and Cases* (2 articles). Considering all the cited references of each article, the most cited journal for MCs articles is *Education and Information Technologies* (223 citations).

- (iii) **Authorship of micro-credentials articles:** A total of 122 authors have contributed to the 56 reviewed articles, with 8% contributing to two or more papers. Newby, T.Z (5 articles), Cheng, Z (3 articles) and Mah, D.K (3 articles) are the three most prolific authors. The contributing authors are affiliated with educational institutions in 22 countries led by the United States (24 articles), Australia (13 articles), Canada (5 articles), and Germany (3 articles). The co-authorship between countries shows four clusters: the United States, Australia, Germany and Saudi Arabia; the Netherlands and Jamaica; Portugal and Lithuania; and China and Taiwan.
- (iv) **Methodology trends of micro-credentials research:** The MC articles in our literature pool have used various research methodologies: empirical, conceptual, review, and mixed method studies (see Fig. 4). Our analysis shows that MCs research is dominated by the use of conceptual and empirical methodologies. Among empirical research, most studies use case study (12 articles), survey (11 articles), or mixed method (case study and survey) (5 articles). None of the articles were based on modelling approach.
- (v) **Theoretical rigor of micro-credentials research:** Though conceptual articles make up 39% of the 56 articles reviewed, only four of them employed a theory. Among the empirical papers reviewed, eight used a theoretical lens. In total, 10 different theories have been applied. These theoretical lenses were mainly drawn on to discuss learning, motivation, curriculum design and technology-related



**Fig. 4** Frequency distribution of research methodologies used in MCs research

issues of MCs and DBs in HE research. A summary of the application of the theories is presented in Table 2.

## 4.2 Thematic analysis

The keyword co-occurrence analysis (in Sect. 3.2) resulted in four clusters: micro-credentials and employability; student learning and technology; micro-credential design and implementation; and curriculum design and assessment.

### 4.2.1 Cluster 1: Micro-credentials and employability

Cluster 1 consists of four themes: MCs, market needs, employability and competency development. Articles under this theme discuss MCs in general and consider MCs as a competency-based learning to fulfill the demands of twenty-first century skills.

**Micro-credentials** MC in HE is an innovative learning model (Alamri et al., 2021) that provides competency-based learning (Wheelahan & Moodie, 2021b), MCs has various names, with DB being the most common (237 times in 43 articles), followed by MC (30 times in 12 articles). MCs are awarded following the acquisition of specific knowledge or demonstration of competence by recognised professional bodies or HE institutions, which maintain the conditions for awarding them (European Commission, 2020; Fanfarelli & McDaniel, 2017). DBs signal the accomplishment of a MC (Cheng et al., 2020). Available in a variety of forms including digital pictures, signs and icons, or a medal (Başal & Kaynak, 2020), DBs are awarded in HE for ‘academic credit’ or ‘recognition of attainment’ and provide validation of the achievement of skills and competencies through explicit evidence (Blackburn et al., 2016).

**Table 2** Major theories applied in MC research

| Theory                                                      | Explanation                                                                                                                                                                    | Articles applying theory                         |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| Achievement goal theory (Pintrich, 2000)                    | Explains the context for interpreting the value of using MCs, DBs, and learners' motivation by performing achievement-related tasks                                            | Alt (2021), Zhou et al. (2019)                   |
| Adult learning theory (andragogy) (Merriam, 2001)           | Explains how learners need self-conception and self-direction for online learning                                                                                              | Behney (2019)                                    |
| Disruptive theory (Christensen et al., 2001)                | Describes how MCs can be implemented to create new markets for learners whose needs can't be satisfied by current learning systems                                             | Cheng et al. (2018)                              |
| Goal setting theory of motivation (Locke & Latham, 2015)    | Explains self-regulated learning, where learners pursue a set goal and need to perform to achieve a task                                                                       | Cheng et al. (2020)                              |
| Human capital theory Becker (1962)                          | Describes that people can increase their skills and competency through education, such as MCs                                                                                  | Wheelahan and Moodie (2021a)                     |
| Innovation diffusion theory (Rogers et al., 2014)           | Contents that complexities in learning and teaching through MCs can act as barriers to successful adoption of MCs                                                              | Stefaniak and Carey (2019)                       |
| Normative curriculum theory (Young, 2014)                   | Explains how external stakeholders such as government, industry and professional bodies guide and influence curriculum design in the form of micro-credentialing               | Krause (2020)                                    |
| Operant learning theory (Wong, 2008)                        | Explains a method of <i>learning</i> relating to rewards and punishments for behaviour. Rewards, such as badges and grades, have been used to extrinsically motivate students  | Delello et al. (2018)                            |
| Self-regulated learning theory (Zimmerman & Schunk, 2001)   | Justifies how learners' metacognitive behaviours and motivation guide the personalised learning process in MCs                                                                 | Carey and Stefaniak (2018), Alamri et al. (2021) |
| Self-determination theory of motivation (Ryan & Deci, 2000) | Describes how MCs can enhance learner engagement and motivate individual learners towards their goal-directed learning behaviour through competency, autonomy, and relatedness | Facey-Shaw et al. (2020), Olsson et al. (2015)   |

**Market needs** To stay relevant in the twenty-first century with everchanging demand for job skills in the market, continued upskilling and reskilling is pivotal for employability (Selvaratnam & Sankey, 2021). Dynamic market requirements for graduate skills radically disrupt supply side offerings from HE institutions (Pizarro Milian & Davies, 2020). MCs offer short competency-based industry-aligned units of learning that reduce discrepancies and bridge skillset gaps (Wheelahan & Moodie, 2021b), providing learners with market relevant skills (Perkins & Pryor, 2021; Ruddy & Ponte, 2019).

**Competency** MCs represent competencies achieved through acquiring specific skill mastery within individual courses, i.e. academic competency (Wilson et al., 2016); upskilling or acquisition and development of job/industry focused skills (Chan et al., 2020; Wheelahan & Moodie, 2021b); and competency focused personalised learning (Alamri et al., 2021; Cheng et al., 2018). Acquiring a DB provides evidence-based accomplishment of skills and competencies (Newby & Cheng, 2020; Olcott, 2021; Perkins & Pryor, 2021). Moreover, DBs and related educational technologies, such as learning analytics, can help to foster academic competencies, including time management, learning skills, technology proficiency, self-monitoring, and research skills (Mah & Ifenthaler, 2018).

**Employability** By preparing learners with the “right” skills to enter the job market, MCs enhance employability (Brown et al., 2021; Miller et al., 2020; Selvaratnam & Sankey, 2021). The introduction of MCs in HE institutions help to refine the curriculum and increase the potential to develop employability skills aligned with latest market requirements (Wheelahan & Moodie, 2021a). At course level, MCs are a useful tool to foster particular graduate capabilities needed in a rapidly changing employment market (Miller et al., 2020; Woods & Woods, 2021). MCs allow students to showcase and share their competencies and capabilities with potential employers (Perkins & Pryor, 2021).

#### 4.2.2 Cluster 2: Student learning and technology

The six themes that form Cluster 2 include student, motivation factor, student engagement, learning, delivery, and technology. Together, they describe the necessary conditions and critical factors driving student completion of MCs.

**Student** In MCs-based learning and badge earning, the student is the core stakeholder. Profiling students based on their learning goals and interests beyond course requirements can help identify learning technology models (such as the personalised learning approach) to support student learning (Alamri et al., 2021; Virkus, 2019). The literature finds students view DBs as a useful assessment tool to take control of and pace learning and tackle assignments. Students consider DBs to be authentic and innovative, and representative of their achievements (LaMagna, 2017; Zhou et al., 2019). Badges can thus serve as incentives for goal achievement and may be tailored to students’ learning approaches (Alt, 2021).

**Motivation factor** Motivation related MCs studies focus on extrinsic and intrinsic motivational factors, rewards/awards, punishments, retention, satisfaction, absenteeism, and achievement. Because MCs are mainly offered online in a self-learning environment, student motivation is important to spur learning and to complete the MCs course (Coleman, 2018). Extrinsic focused student motivation relates to grades, badges, and recognition (Newby & Cheng, 2020). Studies indicate visualisation of achievements through the award of a DB spur motivation, provide social recognition, and encourage learner participation/engagement (Dyjur & Lindstrom, 2017; Gibson et al., 2015; Joseph et al., 2021; Olsson et al., 2015; Yıldırım et al., 2016). At different levels, low-performing students are driven by completion-type badges, while high-performing students need mastery-type badges for motivation (Abramovich et al., 2013). Intrinsic motivation focuses on participation for pleasure or satisfaction and authentic experience (Cheng et al., 2020; Facey-Shaw et al., 2020). Learners are considered as conscious proactive individuals with intrinsic motivation (Olsson et al., 2015). The design of goal setting elements in MCs can facilitate learners' intrinsic motivation (Cheng et al., 2018).

**Student engagement** Research on student engagement emphasises that learning and completion of MCs with DB is connected with extrinsic and intrinsic motivation factors. Student engagement is a product of motivation (Facey-Shaw et al., 2020). A motivated learner will exert quality effort and stay engaged to complete the learning tasks or activities that constitute the MC (Fanfarelli & McDaniel, 2017). As these motivated students complete more course assignments and examinations, they acquire more badges (Coleman, 2018). Number of badges earned, as such, can differentiate the engagement levels of motivated learners (Fanfarelli & McDaniel, 2017).

**Learning** The student learning theme is linked with all themes identified in this study. MCs student learning-related research draws on several learning and motivation theories, such as self-regulated learning, adult learning (andragogy), and operant learning theory to explore learning process, learning experience, learning performance, and learning design. As MCs are designed for expanding personalised learning opportunities, they transfer the focus of HE from teacher-centred to learner-centred environments (Alamri et al., 2021). Through online learning technology, MCs require no direct interaction between learners and educators, and many MCs emphasise self-paced learning (Brown et al., 2021). Individual self-learning capacity and motivation are therefore crucial issues for successful completion of MCs (Olsson et al., 2015). There is an ongoing debate on whether extrinsic (focused on grades and tasks) or intrinsic (focused on participation for pleasure or satisfaction) motivation drives self-learning (Cheng et al., 2020; Coleman, 2018).

**Delivery** MCs may be offered face-to-face or in a hybrid mode that blends the synchronous face-to-face interactive delivery with the online self-learning method, (Abramovich, 2016). The MCs literature on teaching and delivery-related issues stresses the importance of establishing the technical support system for integrating the DB and badge icons with the online learning management platform prior to the delivery of MCs (Pothier,

2021), including having learning analytics algorithms to predict student performance and success and to provide students with personalised feedback for improvement (Mah, 2016). Delivery effectiveness is enhanced when student instructions are unambiguously detailed and learning requirements are clearly communicated to students (Delello et al., 2018). Traditional face-to-face support may be also needed to assist some students to complete MCs (Chan et al., 2020).

**Technology** Technology provides a more attractive alternative medium of offering MCs online in HE. While computer and internet technologies provide the infrastructure that enables the learning, delivery, design and management of MCs learning platforms (Selvaratnam & Sankey, 2021), learning analytics, digital technology, blockchain, and learning management systems are singled out as critical drivers (West & Lockley, 2016). An effective learning management systems (LMS) design incorporating learning analytics algorithms, for instance, can provide learners with personalised feedback and guidance to succeed (Mah, 2016; Mah & Ifenthaler, 2020). Learning analytics technology provides the capability to validate the accuracy of educational records, linking them to the degree or qualification certification system to enable tasks like transfer of course credits and student identification (Krause, 2020). Internet and blockchain technologies facilitate the transmission of micro-credentialing curricula (Ralston, 2021). Only a few articles mentioned blockchain technology in MC offerings. Blockchain has the potential to offer authenticity to MCs with certified digital document verification as well to provide security solutions for third-party transcript services (Ralston, 2021).

#### 4.2.3 Cluster 3: Micro-credential design and implementation

Cluster 3 deals with the implementation issues of MCs, the role of industry partners, the introduction of MCs in HE institutions, and the MC design, review and quality assurance process. Almost all studies in this cluster underline the significance of MCs and DBs in HE and highlight challenges relating to successful implementation.

**Implementation** The implementation of MCs in the HE curriculum is rising, partly fuelled by the rapid move to online teaching and learning. However, MCs and the use of DBs face mounting implementation challenges (Hartnett, 2021): acceptance by market, industry, and employer (Miller et al., 2020; Stefaniak & Carey, 2019); readiness of HE institutions to offer and integrate MCs within the HE curriculum (Hartnett, 2021; Reid & Paster, 2016; Ruddy & Ponte, 2019); readiness of technology and related logistical capabilities (Mah, 2016; Newby & Cheng, 2020); development of student awareness and perceptions (Baker, 2020; Dyjur & Lindstrom, 2017; Reid & Paster, 2016); and effective collaborative partnership between the HE sector and employers (Perkins & Pryor, 2021).

**Industry** Industry partners and employers play a vital role in MCs-based learning, skill development and quality assurance (Brown et al., 2021). Relevant MCs that meet employers' needs have been branded as successful and acceptable (Selvaratnam &



Sankey, 2021). Industry bodies and employers can also offer credentials in-house or in partnership with awarding organisations, such as HE institutions, to set standards and criteria associated with the award of a DB (Eager & Cook, 2020; Miller et al., 2020; Ralston, 2021). Despite calls to engage with human resource recruiters, employers, professional bodies and employer organisations to gauge their acceptance and awareness of digital credentials, limited research has been conducted into employer awareness, acceptance, and the use of DBs in recruitment practices (Perkins & Pryor, 2021).

**Higher education institution** MCs based education is considered an innovative instruction and credentialing strategy for transforming teaching, learning and assessment in HE institutions, and a major disruption in HE with the potential to reduce, if not eliminate, the growing skills gap (Blackburn et al., 2016; Gibson et al., 2016; Noyes et al., 2020). Through digital badging, introducing MCs in HE provides ample opportunities for HE institutions. DBs have been used in HE in a multitude of learning contexts and serve many purposes (Carey & Stefaniak, 2018), not only as a credentialing mechanism but also as an assessment tool to educate and motivate learners (Abramovich, 2016).

**MC design, review and quality assurance** MCs and the use of DBs are still at an early adoption stage (Hartnett, 2021) with little research on the design of MCs support systems (Noyes et al., 2020). When designing MCs it is important to reduce complexity and incorporate different levels of self-efficacy for self-regulated learning to ensure inclusiveness of different categories of learners (Cheng et al., 2020). Equally, educators' capability needs to be considered in designing meaningful MCs to certify the acquisition of generic skills and match with subject learning outcomes and assessments (Alt, 2021). Co-designing MCs with industry to integrate market requirements, employability, and creativity is also essential (Eager & Cook, 2020) to produce a required list of given skills or knowledge proficiency, to offer opportunity to refine, resubmit, and master learning in line with evolving industry needs, and to provide feedback on learning leading to desired levels of competency (Newby & Cheng, 2020). Having the brand of the HE institution itself also speaks to the quality of the MCs; employers may not feel the need to investigate further (Pothier, 2021).

#### 4.2.4 Cluster 4: Curriculum design and assessment

Cluster 4 consists of two themes: assessment, grading and feedback, and course design. Research articles in this cluster explore how MCs have been incorporated into the curriculum.

**Assessment, grading and feedback** Research on assessment, evaluation and feedback of MCs consider how these components are, or should be, provided to learners. MCs, being small in scope, could be completed in shorter time, and can contribute to an academic degree or to an overall assessment strategy (Trepule et al., 2021). Therefore, research studies on this theme have suggested to align MCs with other assessment strategies (Alt, 2021), because poor assessment and evaluation will threaten the validity of MCs, the awarded badges and, ultimately, the reputation of



the awarding body (Wilson et al., 2016). Used as a formative assessment, the award of DBs can help students gauge their progress and likelihood to meet course requirements, identify difficulties and explore means of improvements (Yıldırım et al., 2016). Educators face challenges in providing assessment and feedback to students. Detailed but clear instructions are needed to help students navigate content and improve learning (Besser & Newby, 2019, 2020). Other challenges relate to maintaining a transparent standard for building trust, recognition and quality assurance amongst stakeholders (Brown et al., 2021).

**Course design** For competency-based education based upon the award of a DB, a competency-based curriculum should be in place at program level, course level, and course assessment level, to align with the learning outcomes of evidence-based and competency-focused assessments (Blackburn et al., 2016). Universities can face significant challenges in integrating DBs into their learning and teaching curriculum (Wilson et al., 2016). MCs work best when they are not treated as an ‘add on’ but are carefully designed and aligned with the university curriculum (Gibson et al., 2016).

## 5 Findings and discussion

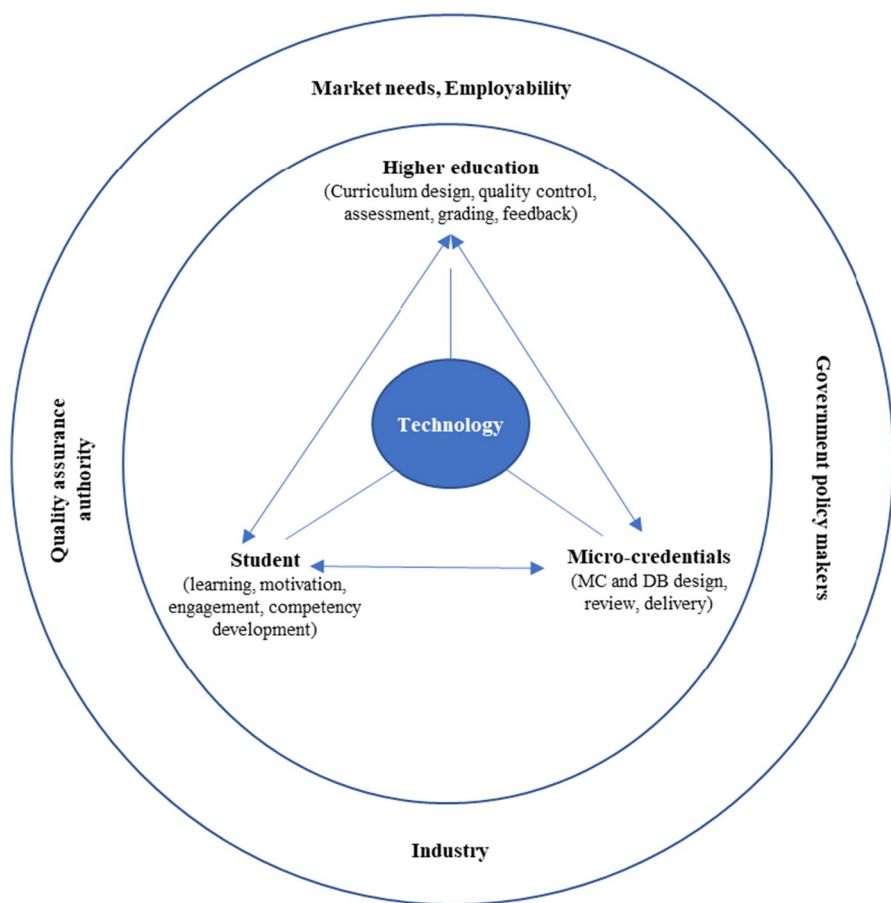
### 5.1 Micro-credential implementation in higher education

Our analysis from the MCs literature reveals that introducing MCs as an integral part of the HE curriculum requires an understanding of the forces at work in two domains. The first domain relates to MCs conceptualisation, development, and operations. The domain is anchored by technology, which drives the platform for MCs and DB design and delivery and acquisition of market-relevant knowledge and skills (Alamri et al., 2021; Mah & Ifenthaler, 2018; Selvaratnam & Sankey, 2021). We label this domain the operational domain.

The second domain encompasses the job market environment and the policy framework governing education provisions. The main forces within this domain include job market skill requirements, contemporary and evolving industry professional practices, and government policies on education quality and standards (Brown et al., 2021; Hartnett, 2021; Miller et al., 2020; Perkins & Pryor, 2021; Stefaniak & Carey, 2019). This second domain forms the MCs eco-system. We depict these two domains in the form of two concentric circles, pointing out the main actors and their roles within each domain based on our synthesized findings from the cluster analysis of key themes (see Fig. 5).

#### 5.1.1 The operational domain

MCs are a technology-based, innovative pedagogical learning tool (Cheng et al., 2020). Technology provides the infrastructure and forms the digital platform for the planning, development and implementation of MCs and DBs; it is an avenue to develop innovative, self-paced, and quality learning (MicroHe, 2019). Technology-enabled MC learning platforms coupled with secure digital certificates (such as DB)



**Fig. 5** A conceptual framework for implementing MCs in HE

enable data portability, trust and knowledge sharing, which can facilitate reliable adoption of MCs in HE (Alamri et al., 2021; Ralston, 2021; Selvaratnam & Sankey, 2021). Learning management systems incorporating learning analytics can reduce an instructor's workload to collect student activity and performance (Alamri et al., 2021; Mah, 2016). These disruptive forces of HE 4.0. with the development of new technologies will require (re)training, reskilling and/or redeployment of HE instructors for technology capabilities (Brown et al., 2021). Technology supports self-learning and integrates learning with HE assessment (Alamri et al., 2021). Students interacting with learning technologies typically prefer easy-to-navigate technologies that can help them discover related content with less effort (Coleman, 2018; Selvaratnam & Sankey, 2021).

HE institutional strategies and policies influence the implementation of MCs (Alt, 2021). Policies and practices for MCs implementation include curriculum and assessment design, MCs design and quality control, instructions for delivery and assessment, and grading (Besser & Newby, 2019; Gibson et al., 2016). In a

dynamically competitive, technologically embedded market environment, new challenges and demands are constantly emerging. HE institutions need to continuously evaluate their learning environments and MC-based assessment processes in order to renew their relevance, and to differentiate themselves from the market and their competitors (Mah & Ifenthaler, 2020; MicroHe, 2017).

As MCs are designed for students, the inclusion of students' voice is therefore essential for the effective implementation of MCs. Research investigating factors which motivate personalised learning, self-learning and technology-assisted-learning finds learner intention and behaviour to be most important (Alamri et al., 2021; Carey & Stefaniak, 2018; Dyjur & Lindstrom, 2017; Facey-Shaw et al., 2020). Competency development and employability are significant external motivational factors that drive students to learn through MCs and complete assessment requirements and attain DBs (Blackburn et al., 2016).

### 5.1.2 The micro-credentials eco-system

Influential actors in the MCs eco-system include industry bodies, student recruitment agencies, and educational quality assurance and government authorities (Baker, 2020; Perkins & Pryor, 2021; Ruddy & Ponte, 2019; Stefaniak & Carey, 2019). HE institutions need to work with both external (industry, recruitment agencies and government) (Perkins & Pryor, 2021) and internal (students and instructors) stakeholders to ensure that the design and delivery of MCs and associated DBs meet desired learning outcomes (competencies) and meet industry needs (jobs) (Hartnett, 2021).

Industry, as a core component of the market, can support HE institutions in developing market-relevant MCs to enhance the employability of graduates (Wheelahan & Moodie, 2021a). This is critical as MCs will only thrive if they are relevant for current and future employers (Selvaratnam & Sankey, 2021). Recruiters and employers appear to be interested in a more competency-based recruitment approach using DBs to complement traditional selection and recruitment processes (Perkins & Pryor, 2021). Some high-profile international companies have recruitment strategies that give weight to MCs and DBs in the selection process (Brown et al., 2021). Accreditation authorities and government policy makers can support MC implementation in HE by influencing the design of MC curricula and assessment (MicroHe, 2017).

## 5.2 Potential of micro-credential-based learning

Our review suggest that enormous potential exists in introducing MCs into HE courses and curriculum to fulfil market skills and competency requirements to support learner employability and workforce demand (Chan et al., 2020; Wheelahan & Moodie, 2021b; Wilson et al., 2016). The implementation of MCs provides opportunities for HE institutions to embrace much sought-after personalised and flexible learning opportunities (Lockley et al., 2016), transferring the focus of HE from teacher-centred to learner-centred environments (Alamri et al., 2021). DBs earned through completion of MCs encourage persistent student engagement by recognising students' generic skills, signalling their achievements, and capturing their learning

paths and making their efforts feel rewarded (Alt, 2021; Delello et al., 2018; Mah & Ifenthaler, 2018). This reinforces the efficacy of MCs-based education as an innovative instructional and credentialing model for transforming teaching, learning and assessment in HE, as well as a major disruptive force in HE with the potential to narrow the evolving skills gap (Gibson et al., 2016; Newby & Cheng, 2020; Noyes et al., 2020; Olcott, 2021).

### 5.3 Challenges of implementing micro-credentials in higher education

The rapid shift to online teaching that has intensified during the COVID pandemic has created an opportunity to re-examine curricula and pedagogies. The literature on MCs is developing quickly, and as HE institutions introduce or plan to introduce MCs and DBs into the curriculum, early adopters of MCs can provide learning experiences and help to address challenges of implementing MCs in HE. As indicated in our conceptual framework (Fig. 5), the success of implementing MCs initiatives depends upon many stakeholders, including governments, industry bodies, educators, and students. We present some of the main challenges of implementing MCs in HE below.

#### 5.3.1 Learners' perceptions of micro-credentials

Despite many anecdotal and conceptual demonstrations of value, the effectiveness of using MCs and DBs in improving learning performance is still largely unknown (Newby & Cheng, 2020). Some students view DBs as redundant, distractive and disturbing (Olsson et al., 2015). A negative or mediocre perception of DB amongst students makes the perceived MC value less prestigious compared to a traditional assessment or certificate of completion (Dyjur & Lindstrom, 2017). It is challenging for students to accept badges as an assessment tool, and continuous learner practice is required to prove effectiveness (Zhou et al., 2019). For learners, the self-conception and self-direction needed for online learning can be confronting and may lead to loss of motivation. Thus, developing an effective LMS design to encourage learners' awareness of MCs and DBs is crucial (Delello et al., 2018).

#### 5.3.2 Higher education readiness

HE institutions are not yet fully ready to adopt the changes necessary to integrate MCs into their curriculum (Hartnett, 2021; Miller et al., 2020). This is not surprising given the nascent state of MCs and the effort, resources and administrative commitment required for successful implementation of MCs (Reid & Paster, 2016). For example, instructional support systems and plans are necessary to increase faculty awareness and participation as a means to increase university-wide adoption of MCs (Wilson et al., 2016). The integration of MCs into learning management systems is important for curating all MCs and related DBs issued by an institution, and maintaining authenticity of records including who issued badges and what badges were awarded (Wilson et al., 2016). Moreover, DBs should not be perceived as effective tools for learning in all settings but rather should be used jointly with other assessment strategies (Alt, 2021).

### 5.3.3 Technological readiness

Technological progress has given rise to the use of DBs in the teaching and learning environment (Behney, 2019). As MCs offered through online learning limit direct contact with the instructor, technological support is essential to help learners adapt to the learning system. Learners typically prefer to interact with learning technologies which are easy to use and to discover content (Selvaratnam & Sankey, 2021). Clear and accessible instructions are therefore essential to support learners in using learning technologies (Coleman, 2018). HE technological readiness in the storage, security and administrative management of badges and badge icons is equally important to maintain integrity and authenticity of badges awarded (Newby & Cheng, 2020).

### 5.3.4 Industry support and awareness

There is a general lack of awareness and readiness amongst HE MC stakeholders, including the market, industry bodies, and employers (Eager & Cook, 2020; Miller et al., 2020; Perkins & Pryor, 2021; Stefaniak & Carey, 2019). Stronger partnerships and collaborative works between the HE sector and other stakeholders is pivotal to establish effective digital credentialing systems (Perkins & Pryor, 2021). Continuous engagement with human resources recruiters, employers, professional and industry bodies and employer organisations would help to gauge their awareness and acceptance of MCs and DBs (Perkins & Pryor, 2021).

### 5.3.5 Logistical readiness and capabilities

As is often the case with introduction of new systems, logistical issues abound surrounding the implementation of MCs in HE. DB issuers and developers need to consider how to prove credibility of badges to stakeholders, whilst HE institutions need to consider how to efficiently incorporate MC badges into the curriculum and on a larger scale coordinate to allow badges to transfer between institutions (Pitt et al., 2019). Issues may arise due to an absence of communication between stakeholders, or an unclear delineation of responsibilities associated with the delivery of the final product (Ruddy & Ponte, 2019).

## 6 Research gaps and future research agenda

Our review identifies several research gaps which may lead to, or stimulate interest in, future studies in MC and DB in HE. We note that less research attention has been given to linking MCs with market needs and perceptions, investigating the challenges of technology adoption, industry support, curriculum design and assessment, and student engagement. The identified research gaps and future research agenda are summarised below.

**Research context gap** The co-authorship cluster of existing MCs in HE research shows that most authors are affiliated with developed countries, such as Europe,

North America and Australia, with low representation from Asian, African, and South American countries. However, MC is an emerging trend of learning, and further research is encouraged in other OECD countries and other geographical areas to consider the influence of different contextual and cultural settings.

**Theory-based research gap** Despite the growing interest in MC and related DB research in HE, little research evaluates theory-based explanations of why MCs might be attractive for students' future learning, and what the drivers and challenges of MC implementation in HE are. We find that MC research consists largely of theories of learner behaviour and motivation, mainly falling under themes such as student motivation, learning, and MC delivery. This is mainly due to the dominance of studies focusing on why MCs and DBs are attractive to students studying at HE institutions. Future research could apply other theories in other themes of MC research, particularly to gain a broader understanding of drivers, potentials, and challenges of introducing MCs in HE.

**Research methodology gap** Our analysis shows that research on MCs is dominated by conceptual studies, and that most of the research in the early stages was dominated by conceptual research methodology. More recently, however, there has been a shift in emphasis to case study interview and survey research. As MC research in HE is at an early stage, we can foresee promising opportunities for conducting case study and longitudinal studies. In addition, case study research can be conducted to capture the experiences of early adopters (HE providers) or the perceptions of external stakeholders on the awareness and acceptance of MCs and DBs issued by HE institutions. Other methodologies, such as modelling can be applied to demonstrate impact and success factors of implementing MCs. The use of machine learning or other artificial intelligence-based research tools were not identified in our review. It may be time to employ these tools to predict the types of MCs that will be in demand, and to match the types of MCs with past performance and student needs.

Gaps identified from emerging themes:

- Limited research has been undertaken to assess current market demand of MCs in HE and MC developments in line with changing business dynamics. Equally, few studies have investigated how MCs influence employer recruitment practices (Perkins & Pryor, 2021). More research is encouraged to consider employers' awareness and acceptance of MCs and how achievement of MCs with DB can help graduates secure employment.
- In HE, the introduction of MCs is a disruptive way of learning. With little or no direct interaction between learners and educators in an online environment, motivating learners to remain engaged throughout the learning process to obtain MCs or DBs is challenging (Coleman, 2018; Olsson et al., 2015). However, in self-paced learning, individual learner control and motivation are critical for successful completion of MCs (Olsson et al., 2015). More studies are therefore needed to explore the connection between learning, delivery and instruction of MCs and

motivation and engagement of students. There is an ongoing debate about whether extrinsic (focused on grades and assignments) or intrinsic (focused on participation for pleasure or satisfaction) motivation drives self-learning, which is essential for MCs studies (Cheng et al., 2020; Coleman, 2018). It is also still unclear whether MCs motivate students more when associated with a DB or work outside the classroom, or as a reward for an assignment for which they have already earned credit or a DB (Delello et al., 2018). Therefore, further research is needed to explore how MCs as well as DBs can influence and facilitate student motivation and learning, and to assess the potential and critical challenges of learning through MCs.

- Technology is the enabler of MCs based learning in HE (Brown et al., 2021). Personalisation in learning cannot take place at scale without technology (Alamri et al., 2021). Without supportive learning technology, implementing MCs in HE is impractical. The level of impact that learning technology has on MC implementation in HE will thus be a fruitful area of investigation. There have been relatively few studies that have investigated diffusion and adoption issues of MCs (Stefaniak & Carey, 2019). Early adopters of MCs can provide learning experiences and help to address challenges of implementing MCs. Further research, therefore, can be designed to investigate challenges of introducing MCs in HE through application of the TOE framework, i.e., adopting technological (T), organisational support and readiness (O), and environmental (E) points of view (Tornatzky et al., 1990). Blockchain can be a tool to offer MCs with authenticity through certified digital documents, such as DBs (Ralston, 2021; Selvaratnam & Sankey, 2021). Yet, there has been limited research on investigating diffusion and adoption issues of blockchain technologies in MCs offerings. Additional research can be conducted to explore how blockchain technologies can facilitate MCs in curriculum design and to offer DBs to improve authenticity of the credentials in the job market.
- Little research has been conducted to investigate readiness of HE institutions to make changes that would be necessary to embrace MCs. Partnerships between HE institutions, recruitment agencies, industry, and government and accreditation bodies can drive development and wider acceptance and use of MCs and DBs as a recognition of learner achievement (Brown et al., 2021; Eager & Cook, 2020; Perkins & Pryor, 2021). It is also vital to consider various (internal and external) stakeholder perspectives together to explore potential developments in this area. Future research is important to investigate how HE authorities can work with industry professional bodies to introduce market focused MCs in their degree programmes to address skill gaps in the market.
- It is to be expected, there would be design, logistical, resource and administrative challenges when implementing MCs in HE. HE institutions need to consider MCs as a strategic goal and deploy resources, such as top-level management support, technical capabilities, and funding to drive MC implementation. There is a paucity of research to support the design and implementation of support systems for MCs (Noyes et al., 2020). Future research could focus on curriculum design and how MCs and DBs can be incorporated into course assessment. Further research can be conducted on how quality assurance, academic integrity, student support, and services can be enhanced if HE institutions introduce MCs from third parties.



## 7 Conclusions

MCs are an innovative, disruptive, and alternative form of learning in HE. Our review of 56 relevant academic journal articles offers several major contributions to the emergent literature on MCs and related DBs in HE. First, this study provides a comprehensive review of research on MCs and DBs in HE, which complements and extends the findings of the previous literature review studies (listed in Table 1). We conduct a systematic literature review of peer-reviewed academic articles from a wide-ranging perspective incorporating all research methodologies and include research from all geographic regions and educational disciplines. Second, this study analyses the extant literature and identifies key knowledge areas related to MC and DB implementation in HE (Fig. 2). Third, this research proposes a conceptual framework (Fig. 5) with several important constructs to implement MCs in HE, which offers a theoretical base for future empirical investigations. We conceptualise the MC knowledge areas into two major areas: ‘operational and technology’ and the ‘MCs eco-system’ and highlight the opportunities and challenges within each domain as well as the HE stakeholders critical to ensuring the success of MCs implementation and market acceptability. The study suggests more attention be given to link MCs with market needs, student engagement, industry support and assessment. Fourth, this research identifies some promising research gaps to help successfully implement MCs and DBs into HE offerings.

We recognise that there are limitations in the literature search procedure. The scope of our research is limited to journal articles published in English. Therefore, we acknowledge that we may have overlooked relevant papers published in other languages. However, including papers published in other languages in the systematic review to capture relevant content would be practically challenging. Additionally, our search process might have been slightly narrowed by the use of the ‘title, abstract, keywords’ option and the consideration of only peer-reviewed ‘journal articles’ and ‘book chapters’ (while excluding grey literature). Although this process may be seen as a form of bias, it is a common and recognised rigorous practice in systematic literature reviews (Noyes et al., 2020).

We also confined our search to the Scopus database, supplemented by cross-checking using Google Scholar. The Scopus database only includes ISI-indexed journals and has limited access to papers in certain journals prior to 1996. However, it covers more than 20,000 refereed journals, which is more comprehensive than many other databases, including Web of Science. The inclusion of additional databases may have resulted in additional relevant articles. However, excluding pre-1996 journal articles is unlikely to have affected the results, since MCs in HE is an emerging and new area of research. Finally, we have ensured transparency by following the replicable methods and reporting results according to the widely acknowledged state-of-the-art systematic review methodologies of Tranfield et al. (2003), Denyer and Tranfield (2009), and (Moher et al., 2015).

As HE institutions, learners, and employers are all looking toward embracing MCs to provide alternative competency-based educational and flexible learning opportunities, our systematic literature review on MCs in HE is timely. The study highlights the role of HE institutions in employing MCs to fill many critical skill gaps in industry.



## Appendix

**Table 3** Bibliometric information of 56 MCs in HE articles

| Authors                                    | Title                                                                                                            | Year | Source title                                           | Theory used                    | Research methods     |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------|--------------------------------|----------------------|
| Abramovich S                               | Understanding digital badges in higher education through assessment                                              | 2016 | On the Horizon                                         |                                | Conceptual           |
| Alamri H.A., Watson S., Watson W           | Learning Technology Models that Support Personalization within Blended Learning Environments in Higher Education | 2021 | TechTrends                                             | Self-regulated learning theory | Review               |
| Alt D                                      | Who benefits from digital badges? Motivational precursors of digital badge usages in higher education            | 2021 | Current Psychology                                     | Achievement goal theory        | Empirical-Survey     |
| Baker N                                    | Introduction to the JTL Special Issue on Digital Learning in Higher Education                                    | 2020 | Journal of Teaching and Learning                       |                                | Conceptual           |
| Behney M                                   | Synthesizing a digital badge for chemistry undergraduates                                                        | 2019 | Reference Services Review                              | Adult learning theory          | Conceptual           |
| Besser E.D., Newby T.J                     | Feedback in a Digital Badge Learning Experience: Considering the Instructor's Perspective                        | 2020 | TechTrends                                             |                                | Empirical-Case study |
| Besser E.D., Newby T.J                     | Exploring the Role of Feedback and its Impact within a Digital Badge System from Student Perspectives            | 2019 | TechTrends                                             |                                | Empirical-Survey     |
| Blackburn R.D., Porto S.C.S., Thompson J.J | Competency-based education and the relationship to digital badges                                                | 2016 | Digital Badges in Education: Trends, Issues, and Cases |                                | Conceptual           |

**Table 3** (continued)

| Authors                                                      | Title                                                                                                 | Year | Source title                                                          | Theory used                    | Research methods         |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------|--------------------------------|--------------------------|
| Brown M., Nic Giolla Mhichil M., Beirne E., Mac Lochlainn C  | The global micro-credential landscape: Charting a new credential ecology for lifelong learning        | 2021 | Journal of Learning for Development                                   |                                | Empirical-secondary data |
| Carey K.L., Stefaniak J.E                                    | An exploration of the utility of digital badging in higher education settings                         | 2018 | Educational Technology Research and Development                       | Self-regulated learning theory | Empirical-Case study     |
| Chan B., Wei R., Fetherston C                                | Innovative digital tools in ebp and information literacy education for undergraduate nursing students | 2020 | Journal of Information Literacy                                       |                                | Empirical-Survey         |
| Cheng Z., Richardson J.C., Newby T.J                         | Using digital badges as goal-setting facilitators: a multiple case study                              | 2020 | Journal of Computing in Higher Education                              | Goal setting theory            | Empirical-Case study     |
| Cheng Z., Watson S.L., Newby T.J                             | Goal Setting and Open Digital Badges in Higher Education                                              | 2018 | TechTrends                                                            | Disruption theory              | Conceptual               |
| Coleman J.D                                                  | Engaging undergraduate students in a co-curricular digital badging platform                           | 2018 | Education and Information Technologies                                |                                | Mixed                    |
| DeLello J.A., Hawley H., McWhorter R.R., Gipson C.S., Deal B | Gamifying education: Motivation and the implementation of digital badges for use in higher education  | 2018 | International Journal of Web-Based Learning and Teaching Technologies | Operant learning theory        | Mixed                    |
| Dyjur P., Lindstrom G                                        | Perceptions and Uses of Digital Badges for Professional Learning Development in Higher Education      | 2017 | TechTrends                                                            |                                | Mixed                    |

**Table 3** (continued)

| Authors                                                                  | Title                                                                                                            | Year | Source title                                                                                                 | Theory used               | Research methods |
|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------------------|---------------------------|------------------|
| Eager, B., Cook, E                                                       | Micro-credentialing of entrepreneurship education in a practice-based undergraduate engineering context          | 2020 | Entrepreneurship Education and Pedagogy                                                                      |                           | Conceptual       |
| Facey-Shaw L., Specht M., van Rosmalen P., Bartley-Bryan J               | Do Badges Affect Intrinsic Motivation in Introductory Programming Students?                                      | 2020 | Simulation and Gaming                                                                                        | Self-determination theory | Empirical-Survey |
| Fanfarelli J.R., McDaniel R                                              | Exploring digital badges in university courses: Relationships between quantity, engagement, and performance      | 2017 | Online Learning Journal                                                                                      |                           | Empirical-Survey |
| Garnett T., Button D                                                     | The use of digital badges by undergraduate nursing students: A three-year study                                  | 2018 | Nurse Education in Practice                                                                                  |                           | Empirical-Survey |
| Gibson D., Coleman K., Irving L                                          | Learning journeys in higher education: Designing digital pathways badges for learning, motivation and assessment | 2016 | Foundation of Digital Badges and Micro-Credentials: Demonstrating and Recognizing Knowledge and Competencies |                           | Conceptual       |
| Gibson D., Ostashevski N., Flintoff K., Grant S., Knight E Hartnett, M.K | Digital badges in education                                                                                      | 2015 | Education and Information Technologies                                                                       |                           | Conceptual       |
|                                                                          | How and why are digital badges being used in higher education in New Zealand?                                    | 2021 | Australasian Journal of Educational Technology                                                               |                           | Mixed            |
| Joseph M.A., Natarajan J., Buckingham J., Al Noumani M                   | Using digital badges to enhance nursing students' attendance and motivation                                      | 2021 | Nurse Education in Practice                                                                                  |                           | Empirical-Survey |
| Kim J., Castelli D.M                                                     | Effects of gamification on behavioral change in education: A meta-analysis                                       | 2021 | International Journal of Environmental Research and Public Health                                            |                           | Review           |

**Table 3** (continued)

| Authors                                              | Title                                                                                                            | Year | Source title                                                                                                 | Theory used       | Research methods     |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------------------|-------------------|----------------------|
| Krause K.-L.D                                        | Vectors of change in higher education curricula                                                                  | 2020 | Journal of Curriculum Studies                                                                                | Curriculum theory | Conceptual           |
| LaMagna M                                            | Placing digital badges and micro-credentials in context                                                          | 2017 | Journal of Electronic Resources Librarianship                                                                |                   | Conceptual           |
| Lemoine, P.A., Richardson, M.D                       | Micro-credentials, nano degrees, and digital badges: New credentials for global higher education                 | 2015 | International Journal of Technology and Educational Marketing (IJTEM)                                        |                   | Conceptual           |
| Lockley A., Derryberry A., West D                    | Drivers, affordances and challenges of digital badges                                                            | 2016 | Foundation of Digital Badges and Micro-Credentials: Demonstrating and Recognizing Knowledge and Competencies |                   | Conceptual           |
| Mah D.-K                                             | Learning Analytics and Digital Badges: Potential Impact on Student Retention in Higher Education                 | 2016 | Technology, Knowledge and Learning                                                                           |                   | Conceptual           |
| Mah D.-K., Ifenthaler D                              | What do first-year students need? Digital badges for academic support to enhance student retention               | 2020 | Journal of Applied Research in Higher Education                                                              |                   | Empirical-Survey     |
| Mah D.-K., Ifenthaler D                              | Students' perceptions toward academic competencies: The case of German first-year students                       | 2018 | Issues in Educational Research                                                                               |                   | Review               |
| Miller K.K., St Jorre T.J.D., West J.M., Johnson E.D | The potential of digital credentials to engage students with capabilities of importance to scholars and citizens | 2020 | Active Learning in Higher Education                                                                          |                   | Empirical-Case study |
| Newby T.J., Cheng Z                                  | Instructional digital badges: effective learning tools                                                           | 2020 | Educational Technology Research and Development                                                              |                   | Empirical-Survey     |

**Table 3** (continued)

| Authors                                             | Title                                                                                       | Year | Source title                                                | Theory used               | Research methods     |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------|------|-------------------------------------------------------------|---------------------------|----------------------|
| Noyes J.A., Welch P.M., Johnson J.W., Carboneau K.J | A systematic review of digital badges in health care education                              | 2020 | Medical Education                                           |                           | Review               |
| Olcott D., Jr                                       | Micro-Credentials: A Catalyst for Strategic Reset and Change in U.S. Higher Education       | 2021 | American Journal of Distance Education                      |                           | Conceptual           |
| Olsson M., Mozellius P., Collin J                   | Visualisation and gamification of e-Learning and programming education                      | 2015 | Electronic Journal of e-Learning                            | Self-determination theory | Empirical-Case study |
| Pedro L., Santos C., Aresta M., Almeida S           | Peer-supported badge attribution in a collaborative learning platform: The SAPO Campus case | 2015 | Computers in Human Behavior                                 |                           | Empirical-Case study |
| Perkins J., Pryor M                                 | Digital badges: Pinning down employer challenges                                            | 2021 | Journal of Teaching and Learning for Graduate Employability |                           | Empirical-Survey     |
| Pitt C.R., Bell A., Strickman R., Davis K           | Supporting learners' STEM-oriented career pathways with digital badges                      | 2019 | Information and Learning Science                            |                           | Empirical-Case study |
| Pizarro Milian R., Davies S                         | Forecasting the impacts of the "future of work" on universities: a sociological perspective | 2020 | On the Horizon                                              |                           | Empirical-Case study |
| Pothier W.G                                         | Using digital badges in business information literacy instruction: a case study             | 2021 | Journal of Business and Finance Librarianship               |                           | Empirical-Case study |
| Ralston, S.J                                        | Higher Education's microcredentialing craze: a postdigital-Deweyan critique                 | 2021 | Postdigital Science and Education                           |                           | Conceptual           |
| Reid A.J., Paster D                                 | A case study of digital badges in composition courses                                       | 2016 | Digital Badges in Education: Trends, Issues, and Cases      |                           | Conceptual           |

**Table 3** (continued)

| Authors                                                                | Title                                                                                                                                    | Year | Source title                                                                                                 | Theory used                 | Research methods     |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------|
| Ruddy C., Ponte F                                                      | Preparing students for university studies and beyond: a micro-credential trial that delivers academic integrity awareness                | 2019 | Journal of the Australian Library and Information Association                                                |                             | Conceptual           |
| Selvaratnam R.M., Sankey M.D                                           | An integrative literature review of the implementation of microcredentials in higher education: Implications for practice in Australasia | 2021 | Journal of Teaching and Learning for Graduate Employability                                                  |                             | Review               |
| Stefaniak J., Carey K                                                  | Instilling purpose and value in the implementation of digital badges in higher education                                                 | 2019 | International Journal of Educational Technology in Higher Education                                          | Innovation diffusion theory | Empirical-Case study |
| Trepule E., Volungevičienė A., Teresevičienė M., Greenspon R., Costa N | How to Increase the Value of Digital Badges for Assessment and Recognition in Higher Education. A University Case                        | 2021 | Informatics in Education                                                                                     |                             | Empirical-Case study |
| Virkus S                                                               | The use of Open Badges in library and information science education in Estonia                                                           | 2019 | Education for Information                                                                                    |                             | Conceptual           |
| West D., Lockley A                                                     | Implementing digital badges in Australia: The importance of institutional context                                                        | 2016 | Foundation of Digital Badges and Micro-Credentials: Demonstrating and Recognizing Knowledge and Competencies |                             | Conceptual           |
| Wheelahan L., Moodie G                                                 | Analysing micro-credentials in higher education: a Bernsteinian analysis                                                                 | 2021 | Journal of Curriculum Studies                                                                                | Human capital theory        | Conceptual           |
| Wheelahan L., Moodie G                                                 | Gig qualifications for the gig economy: micro-credentials and the 'hungry mile'                                                          | 2021 | Higher Education                                                                                             |                             | Conceptual           |

**Table 3** (continued)

| Authors                                     | Title                                                                                                                                   | Year | Source title                                                                                                 | Theory used             | Research methods     |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------------------|-------------------------|----------------------|
| Wilson B.G., Gasell C., Ozyer A., Scrogan L | Adopting digital badges in higher education: Scoping the territory                                                                      | 2016 | Foundation of Digital Badges and Micro-Credentials: Demonstrating and Recognizing Knowledge and Competencies |                         | Empirical-Case study |
| Woods K., Woods J.A                         | Less Is More: Exploring the Value of Micro-Credentials Within a Graduate Program                                                        | 2021 | Journal of Continuing Higher Education                                                                       |                         | Conceptual           |
| Yıldırım S., Kaban A., Yıldırım G., Çelik E | The effect of digital badges specialization level of the subject on the achievement, satisfaction and motivation levels of the students | 2016 | Turkish Online Journal of Educational Technology                                                             |                         | Mixed                |
| Zhou L., Chen L., Fan Q., Ji Y              | Students' perception of using digital badges in blended learning classrooms                                                             | 2019 | Sustainability (Switzerland)                                                                                 | Achievement goal theory | Empirical-Survey     |



**Abbreviations** MC: Micro-credential; HE: Higher education; DB: Digital badge

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**Data availability** Bibliometric data generated for the 56 selected articles are included in an Appendix Table 3.

## Declarations

**Conflict of interest** All authors declare that they have no conflict of interests associated with this submission.

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